Atmosphere TV

DOOH Venue Incrementality & Media-Mix Measurement

A causal measurement framework, calibrated media-mix model, budget optimizer, and venue revenue + retention models — built to answer the questions Atmosphere's advertising business and its own network economics run on.

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How I'd approach this role

Before the demo — the operating thesis behind it: what this role should do for Atmosphere, and how I'd prioritize the first few months as its first data scientist.

Advertising-side measurement

Prove Atmosphere's own incremental foot traffic with methods advertisers can trust — the sell-side differentiator this business runs on. (Answers Q1–Q2.)

Venue-network health

Know which venues are generating less ad revenue than their audience is worth, and which are at risk of leaving — protect and grow the inventory Atmosphere actually sells. (Answers Q3–Q4.)

Not a “quick win vs. foundation” tradeoff — do both in one move

A stratified RCT geo-holdout is the fastest trustworthy result available — and it doubles as the calibration anchor every other method here relies on: synthetic control's placebo check, the MMM's scale, and the venue models' causal-value feature. Ship it first and the “quick win” and the “foundation” are the same deliverable. The venue-network data pipeline (economics + retention features) doesn't have to wait on it — it builds in parallel.

What “done” means

Sales and account teams making different decisions with it: a trusted lift number in a client pitch, a budget conversation anchored to a real response curve, an outreach list ranked by value × risk instead of instinct — not just a model with a good accuracy number.

What follows is one concrete example of applying this approach — a synthetic demo.

Honest scope

Before the demo that follows: the scope caveats specific to this project, stated up front rather than saved for the closing slide.

Synthetic data

Every figure in this project is synthetic. The focus here is the methodology for solving the problem, not real company data or figures.

Media-effectiveness & rate-card figures

Dwell time, screen count, ad rate card, and similar parameters shown are illustrative demo values, not researched real industry benchmarks. In production: Nielsen OOH, DSP data (e.g. Vistar), or Atmosphere's own play logs and rate card.

Multi-touch attribution — deliberately not built

Atmosphere's ambient-screen model has no individual-level, cross-venue touchpoint log by default. Building MTA would require purchased mobile location/device-matching data — a real but non-default assumption, so it's scoped out rather than forced.

Online-purchase attribution — also out of scope

Every method here measures incremental foot traffic, not downstream online purchases. Tying exposure to e-commerce conversions needs a device-matched exposure-to-transaction panel — a further non-default assumption layered on top of MTA's.

Cost & revenue assumptions

The $/frequency-unit figures behind the budget allocator, and the ad-rate-card behind the venue-revenue model, are illustrative, editable placeholders — in production these come from Atmosphere's own rate card by venue type and daypart.

Retention model — its own caveats

Churn-hazard, competitive-geo, and engagement-signal parameters are illustrative, not researched attrition benchmarks. geo_cluster is deliberately excluded (small-sample tradeoff); realized_ad_revenue's importance is a venue_type confound, not a causal driver; no prospect-side retention model.

Four questions this project answers

Q1–Q2 answer the advertising side (what Atmosphere sells); Q3–Q4 answer the venue-network side (what Atmosphere runs).

1

Did the campaign actually work?

Isolate the TRUE incremental foot traffic caused by ad exposure — net of seasonality, trend, and each venue's own baseline pattern.

The sell-side differentiator: measurement Atmosphere's go-to-market team can take to market and clients can trust.

2

How should budget be spent?

Given a budget an advertiser has already committed to Atmosphere, how should it split across restaurants, gyms, bars, and waiting rooms — accounting for each venue type's own diminishing-returns curve?

An allocation Atmosphere optimizes on its own inventory and hands advertisers as part of the media plan — not a cross-channel budget call made elsewhere.

3

Where's the revenue upside?

Which existing venues are under-monetized relative to their own traffic and quality — and which prospective venues are worth prioritizing for expansion?

Atmosphere's own buy-side value question.

4

Which venues are at risk?

Which existing venues are likely to leave the network, and which ops-visible signals predict it — combined with Q3's revenue into one value-×-risk priority.

Atmosphere's own buy-side risk question.

Design: one pipeline, confidence flows from experiment to product

1 — Q1

RCT geo-holdout

High confidence

Randomized treated/holdout venues. Balance directly verified.

2 — Q1

Synthetic control

Moderate confidence

Covers historical, non-randomized campaigns. Validated with placebo tests.

3 — Q2

MMM, RCT-calibrated

Shape + scale

Adstock/saturation shape from aggregate data; scale pinned by the RCT.

4 — Q2

Budget allocator

Exact DP

Multiple-choice knapsack over calibrated response curves.

5 — Q3

Venue revenue model

GBT, honest OOF

Venue characteristics + the calibrated lift (steps 1–3) as a feature. Flags under-monetized venues; ranks expansion prospects.

6 — Q4

Venue retention model

GBT classifier, honest OOF

Ops-visible signals (engagement, uptime, complaints, outreach) + realized revenue. Flags at-risk venues; combines with Q3 into one value×risk priority.

Answers Q1 — did the campaign work?

Answers Q2 — how should budget be spent?

Answers Q3 — revenue upside?

Answers Q4 — retention risk?

Steps 1–2 answer Q1: RCT for randomized campaigns, synthetic control extending coverage to historical, non-randomized ones. Steps 3–4 turn that into Q2: MMM shapes the response curve, the RCT pins its scale, and the allocator solves the exact split. Step 5 answers Q3: the venue-revenue model reuses the calibrated per-exposure lift from steps 1–3 as a feature. Step 6 answers Q4: a separate churn classifier flags at-risk venues, then combines its risk score with step 5's revenue into one value-×-risk priority — one connected system feeding all four answers, not four separate projects bolted together.

Two data sources: a designed holdout test, and campaign history

4 venue types × 100 venues each, 104 weeks (52 pre-period + 52 campaign)

RCT pool

Randomly split treated / holdout within venue_type × geo_cluster × traffic-tier strata for a fixed benchmark campaign. This is a real Atmosphere-designed measurement product, not a historical campaign.

Used to anchor confidence — assignment is verified, not assumed.

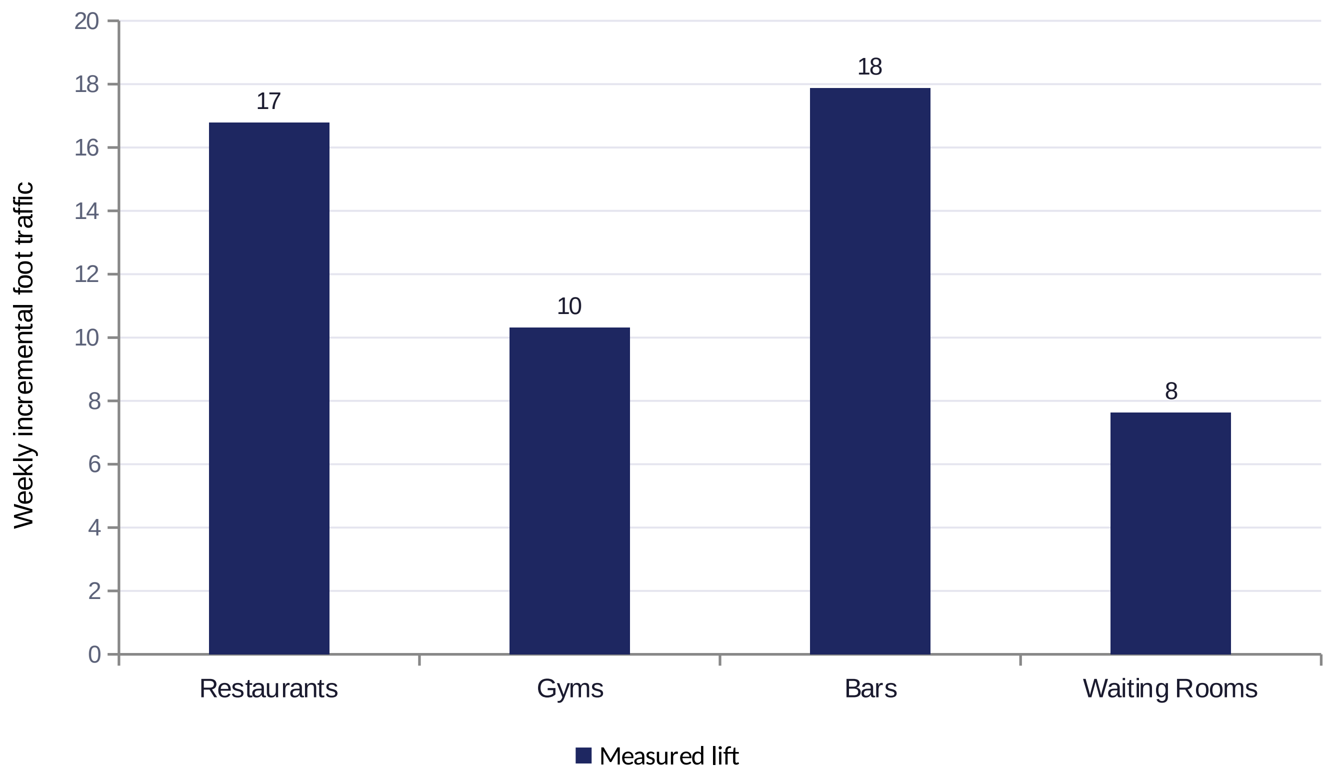
Observational pool

Advertisers activate venues themselves — non-randomly. Higher-baseline-traffic venues are systematically more likely to be activated (a real selection mechanism), and weekly frequency varies venue-to-venue.

A naive before/after comparison here is confounded by selection — this is what synthetic control and the MMM have to work around.

The proof point: ads move real foot traffic

2/16 balance tests sig.



Venue type	95% CI	p-value
Restaurants	[11.1, 22.4]	< 0.001
Gyms	[5.2, 15.4]	< 0.001
Bars	[13.3, 22.5]	< 0.001
Waiting Rooms	[5.1, 10.2]	< 0.001

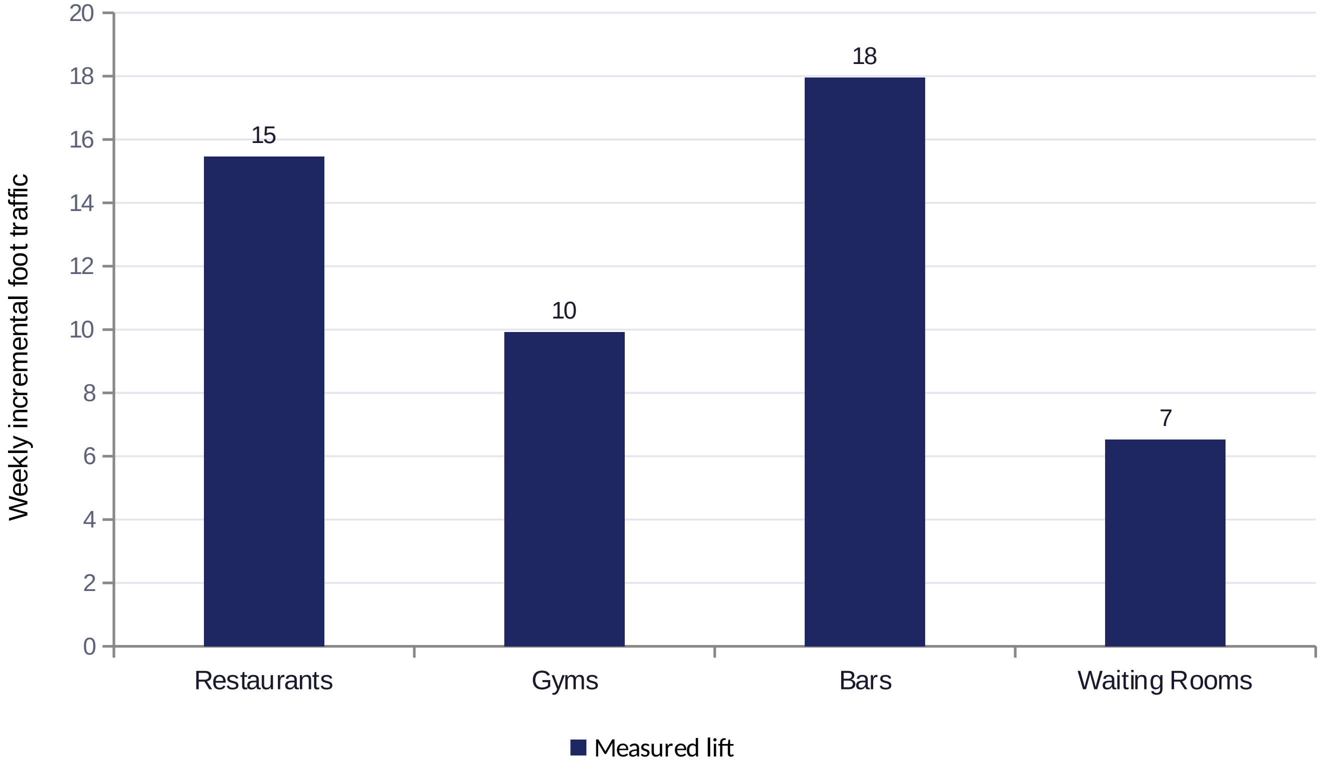
Why 2/16 isn't a red flag

16 tests = 4 pre-treatment covariates (baseline traffic level, dwell time, screen count, audience quality) × 4 venue types. Only 2 came back significant at p≤0.05.

If randomization is clean, the count of false positives across 16 independent tests follows Binomial(16, 0.05): mean 0.8, P(≥2) ≈ 19%. Two sits well inside that distribution's main mass — nowhere near a red-flag tail value like 8+. Randomization worked as designed, not assumed.

Measuring the campaigns we couldn't randomize

Most advertisers' campaigns already ran, on venues they picked — not us. Synthetic control reconstructs the counterfactual anyway, with its own built-in checks.

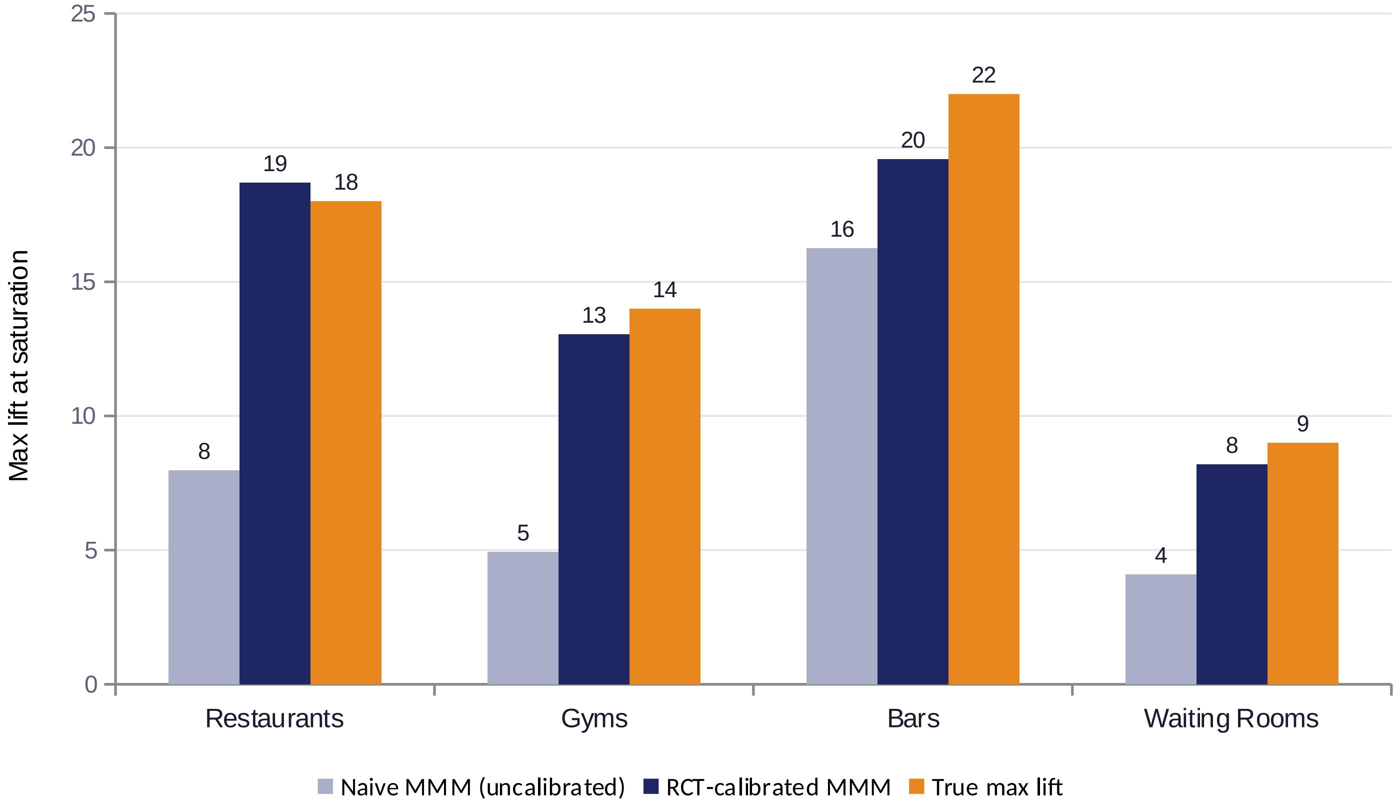


Venue type	Placebo p	Pre-RMSPE
Restaurants	0.160	26.4
Gyms	0.050	14.4
Bars	0.100	17.8
Waiting Rooms	0.110	8.3

Two validation checks, since synthetic control has no closed-form standard error: pre-period fit quality (RMSPE — poor fits are flagged and excluded from the aggregate estimate) and an in-space placebo test on every donor venue.

MMM: an uncalibrated model would have underpriced results

Adstock + saturation shape comes from the aggregate weekly series; scale is pinned by the RCT's high-confidence estimate.



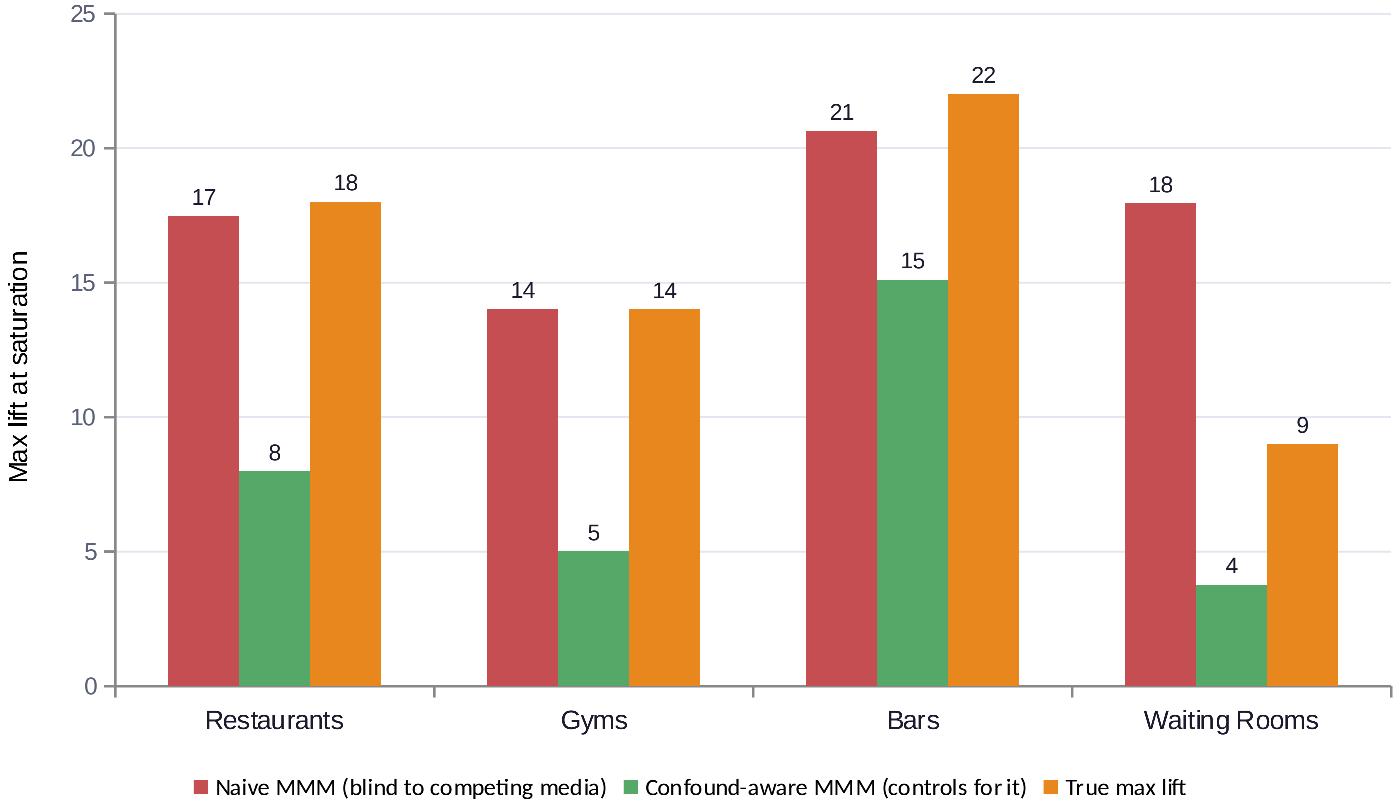
Calibration adjustment

Restaurants	+134%
Gyms	+165%
Bars	+20%
Waiting Rooms	+100%

For 3 of 4 venue types, the naive observational fit understated true incremental lift by 100–165% — the kind of gap real MMM practice cites as its core identification problem, and exactly why the RCT anchor matters.

Robustness check: does competing media break this MMM?

A synthetic “other advertisers were busy too” shock, correlated with Atmosphere's own exposure ramp-up — layered onto the validated data as a labeled overlay, not a change to it.



RCT estimate shift under the identical shock

0.00

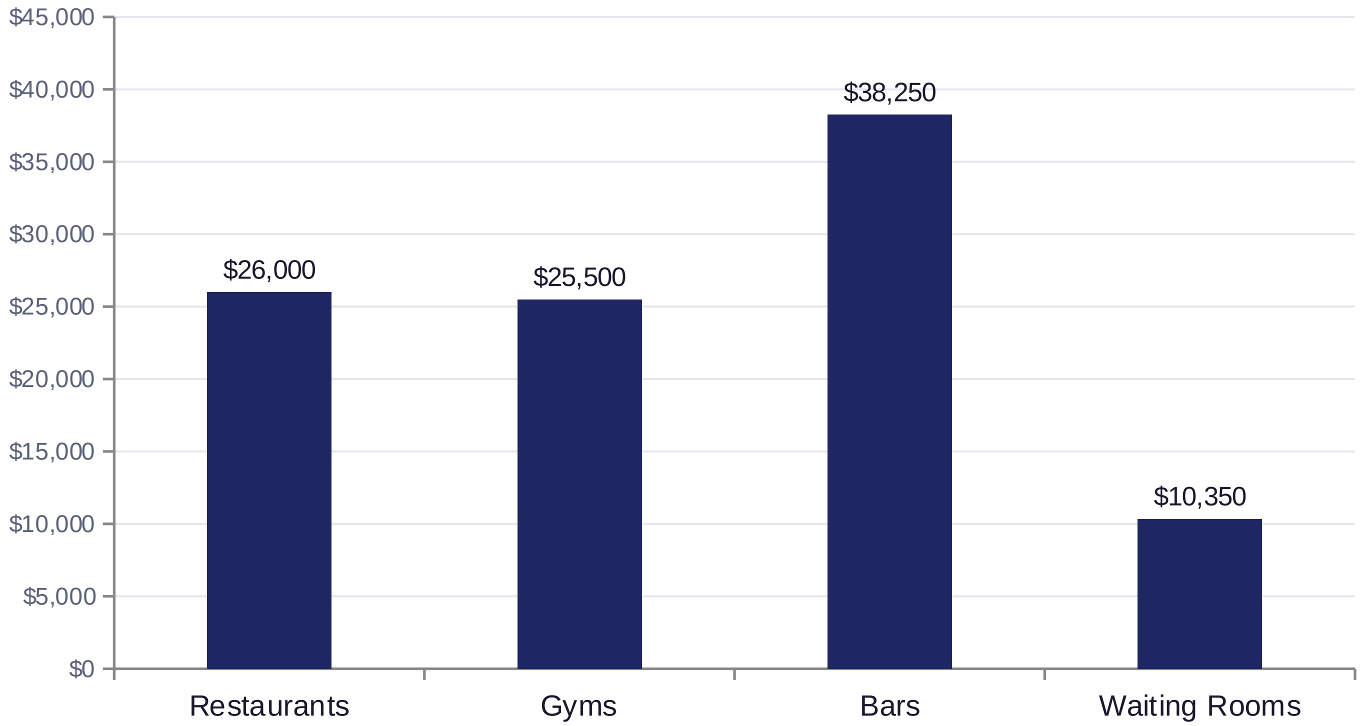
every venue type — the shock hits both arms equally, so a randomized comparison is mechanically immune to it

Naive MMM, by contrast, swings from useful to actively misleading: for 3 of 4 venue types the confound-blind fit was off by 2–4× — even though the injected correlation between competing media and Atmosphere's own exposure ramp is only 0.35 (moderate, not a worst case). Controlling for the confound (green) recovers this model's own uncalibrated baseline from the previous slide — still short of true max lift (orange) until the RCT-scale calibration from that same slide is applied on top.

Why this is scoped this way: deciding how an advertiser splits budget across Atmosphere + TV + social is the advertiser's/agency's call — Atmosphere has no visibility into competitors' channel data anyway. What Atmosphere's DS role should own is netting competing media OUT of its own causal read (the JD's own phrasing), which the RCT does by design and a single-channel MMM only does with an explicit control.

Budget allocator: exact DP, not a greedy walk

A Hill/S-shaped response curve is CONVEX before its inflection point — a greedy "spend the next dollar on whichever venue type currently looks best" heuristic isn't guaranteed optimal there. An earlier greedy version of this allocator actually underperformed a naive equal-split baseline; the DP formulation below has no concavity requirement and is guaranteed to find the grid-optimal allocation.



$100,000 weekly budget example

$20,000 budget **+48.6%**

vs. naive equal-split

$50,000 budget **+11.1%**

vs. naive equal-split

$100,000 budget **+2.4%**

vs. naive equal-split

The optimizer's edge is largest when budget is scarce — exactly when allocation decisions matter most.

The other side of the business

One system answers the advertising side. The same system answers the venue-network side too.

Advertising incrementality

- Prove and price advertising incrementality
- Feeds go-to-market: sell-side differentiator, client trust
- RCT + synthetic control + calibrated MMM + budget DP

Feeds the venue-economics model as a validated causal-value input (calibrated per-exposure lift), not a separate silo.

Venue network economics

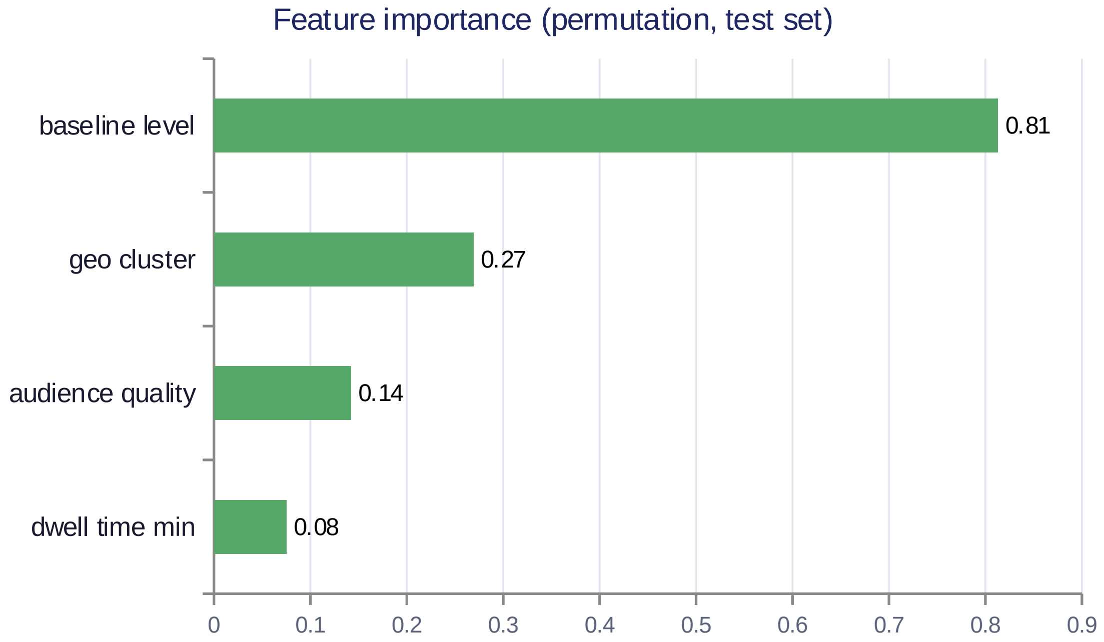
- Predict realized ad-revenue (Q3) and 90-day churn risk (Q4) from characteristics and ops signals
- Flag under-monetized (Q3) and at-risk (Q4) existing venues for sales/ops follow-up
- Rank prospective venues for expansion priority (Q3)
- Combine Q3 revenue + Q4 risk into one value-×-risk priority quadrant
- GBT regressor + classifier, honest out-of-fold evaluation throughout

"Smarter on both sides of the business" — the venue-side model shares its data foundation and causal-value input with the advertising side, rather than being a disconnected second project. Results on the next slide.

Predicting venue ad-revenue, honestly

Gradient-boosted trees on observable venue characteristics + the calibrated per-exposure lift as a feature.

0.84	**22.1%**	**0.94**	**90% vs 28%**
Held-out R^2	Held-out MAPE	Corr. w/ latent true potential	Flagged-tail latent gap rate



Top under-monetized venues (of 20 flagged)

Venue #85 (Restaurants)	-62%
Venue #309 (Waiting Rooms)	-61%
Venue #53 (Restaurants)	-61%
Venue #247 (Bars)	-60%

Top prospect venues (expansion priority)

P0018 — Bars (existing market)	$684
P0055 — Restaurants (existing market)	$524
P0000 — Bars (new market)	$500
P0001 — Bars (new market)	$496

Flags come from 5-fold out-of-fold predictions, never a model scoring the venue it was trained on. The calibrated-lift feature carries near-zero importance — honest, since it's constant within venue_type once venue_type itself is a feature.

Predicting venue churn risk, honestly

Gradient-boosted classifier on ops-visible signals (engagement, uptime, complaints, competitor outreach) — the other half of "what makes a venue valuable and what puts it at risk."

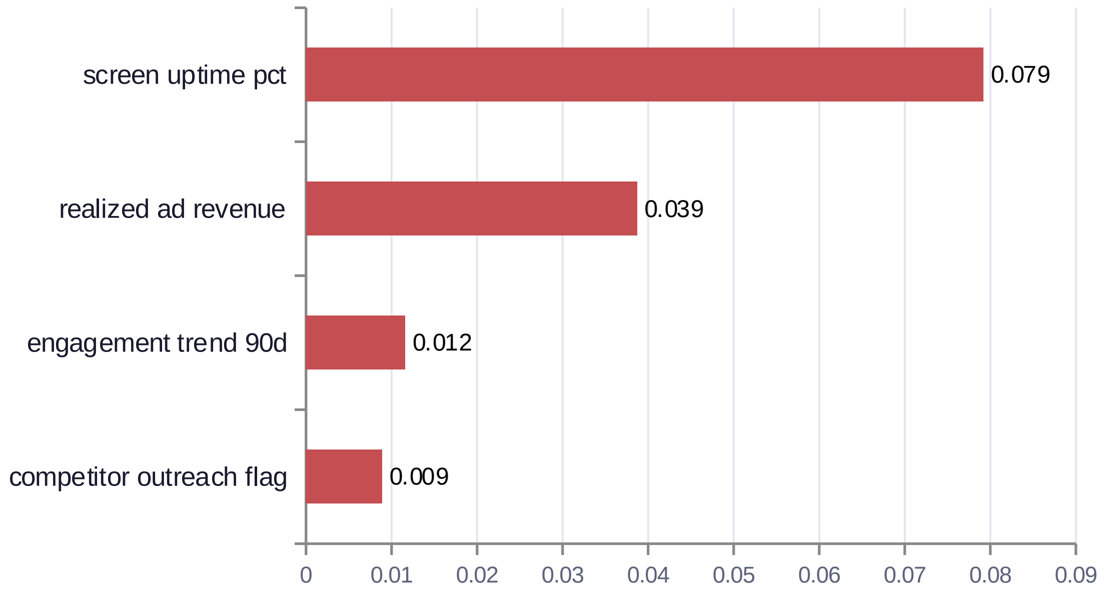
0.65
Held-out AUC (oracle ceiling: 0.73)

0.27
Held-out PR-AUC

0.73
OOF corr. w/ latent true risk

100% vs 31%
Flagged-tail latent risk rate

Feature importance (permutation, test set) — leading indicators



Top at-risk venues (of 20 flagged, by OOF risk)

Venue #281 (Bars)	60%
engagement declining → content/placement review	
Venue #200 (Bars)	55%
engagement declining → content/placement review	
Venue #12 (Restaurants)	52%
uptime issue → dispatch technical support	

Priority quadrant (Q3 revenue × Q4 risk)

Save now (high value, high risk)	136
Monitor (low value, low risk)	136
Protect (high value, low risk)	64
Low priority (low value, high risk)	64

Flags come from 5-fold OOF predictions; AUC/PR-AUC read modest next to the revenue model's R^2 only because the oracle ceiling here is 0.73 AUC. realized_ad_revenue's importance is a venue_type confound, not a causal driver. Each "Save now" / flagged venue also carries a SHAP top driver (from the fold model that never saw it) mapped to a concrete action — technical dispatch, AM retention call, content review — not a generic "reach out."

Takeaways

1. Match the method to the assignment mechanism — RCT where randomization is designed, synthetic control where it isn't, each with its own validation check.

2. An uncalibrated MMM understated true incremental value by up to 165% here — experiment calibration isn't optional polish, it's the identification fix.

3. Optimization needs the right algorithm for the curve's shape — a greedy heuristic silently lost to a naive baseline on non-concave response curves; the exact DP formulation doesn't.

4. One connected system, not four separate projects: the venue-revenue model reuses the causal pipeline's per-exposure value as an input; the retention model then combines its own churn-risk score with that revenue prediction into one value-×-risk priority quadrant — both models recover their respective latent ground truths (0.95 and 0.73 correlation) before either is trusted.

Thank you